GE Medical Systems

Technical Publications

GEMS Part Number 2321162

Revision 1

TEAL PDU Module in ACGD LITE

Operating Documentation

TABLE OF CONTENTS

IMPORTANT SAFETY INSTRUCTIONS

SAVE THESE INSTRUCTIONS – This manual contains important instruction for the ACGD LITE Power Distribution Unit (PDU) that must be followed during installation, operation, and maintenance.

DANGER

Electrical shock hazard. Opening enclosure exposes hazardous voltages. Refer service to qualified personnel.

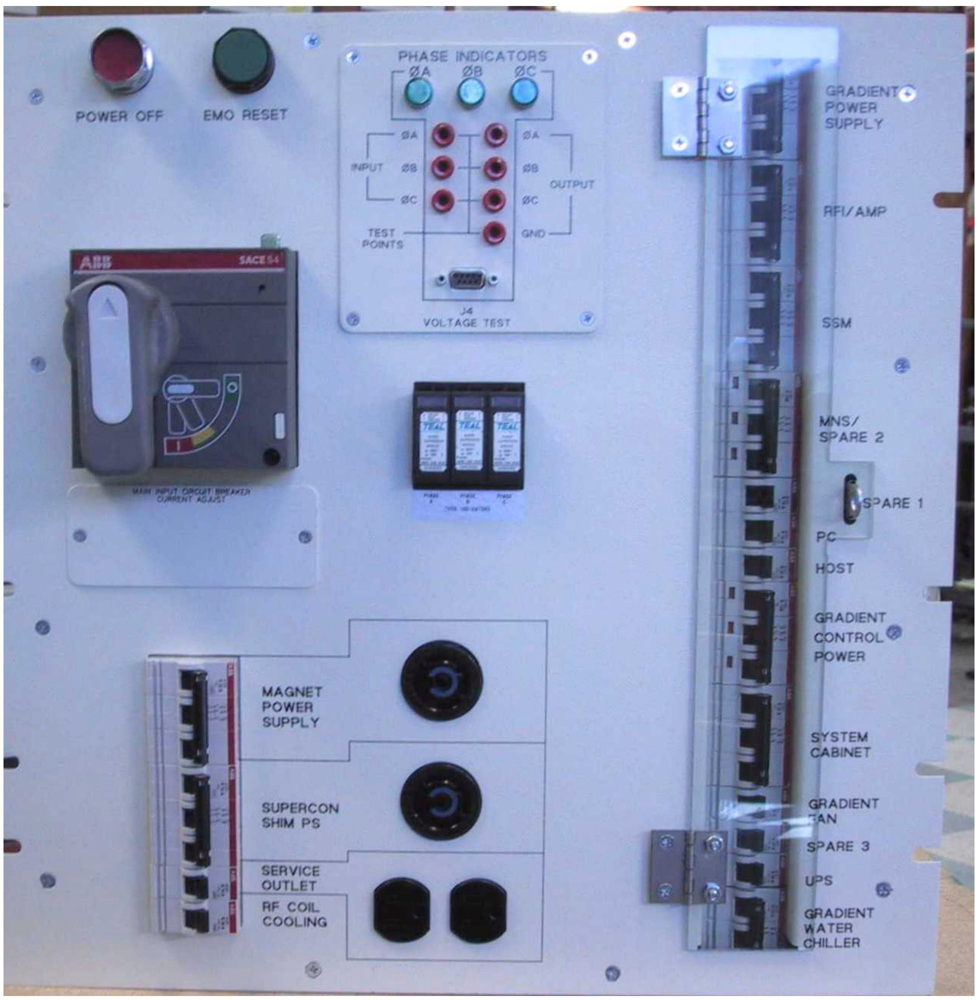
IMPORTANT

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

1- INTRODUCTION

1-1 General Description

The ACGD LITE Power Distribution Unit (PDU) distributes power to various components of a General Electric Medical Systems MRI system while providing surge suppression and electrical isolation for the power line. Three phase input voltage is selectable among six levels: 200/208/380/400/415/480 VAC. Rated frequency is 50/60 Hz. Power rating is 35 KVA continuous, 42 KVA instantaneous. The PDU mounts in a standard 19-inch rack.



Power Distribution Unit
Illustration 1-1

1-2 Scope

This manual describes procedures for installation and basic operation of the ACGD LITE Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

SPECIFICATIONS

Specifications subject to revision without notice

GE Part Number
2321162

Electrical Specifications

Power Rating
35 KVA Continuous; 42 KVA Instantaneous

Input Voltage
200/208/380/400/415/480 VAC Delta, User Selectable (factory wired for 480VAC Delta)

Output Voltage
208/120 VAC WYE

Overload Protection
Main Input Circuit Breaker with adjustable trip settings (factory set to 60A);
Output distribution circuit breakers

Load Current
97.2 Amps total maximum continuous (208V)

Transformer Impedance
Less than 2%

Efficiency
98% Typical

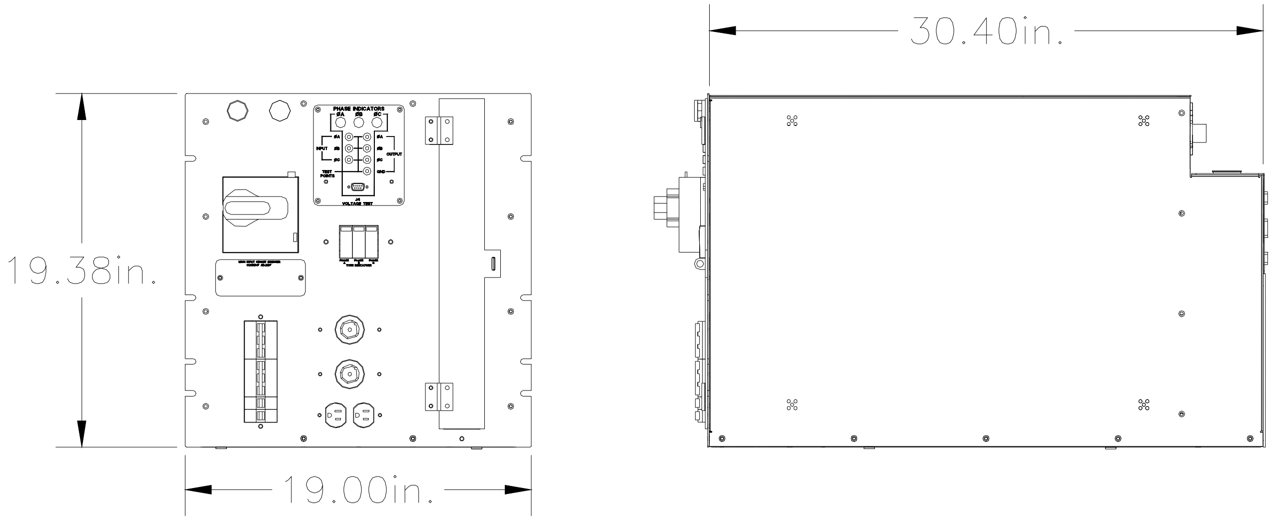
Environmental Specifications

Heat Dissipation
1,876 BTU/Hr

Mechanical Specifications

Weight
600 lbs. / 272 kg

Dimensions
See Illustration 2-1



ACGD LITEPDU Overall Dimensions
Illustration 2-1

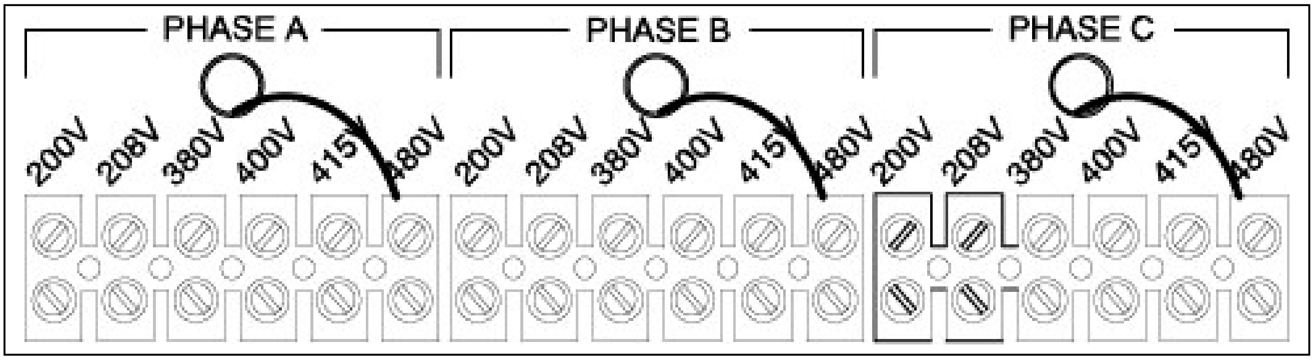
2- INSTALLATION AND OPERATION

3-1 Installation

3-1.1 Input Voltage Selection

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 200, 208, 380, 400, 415, or 480 VAC (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock and tag appropriately.
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, check to be sure that no voltage is being applied to the input of the PDU. Main Input Terminal Board is located on rear panel (refer to illustration 3-5).
3. Remove the cover panel from the front panel of the PDU marked “MAIN INPUT CIRCUIT BREAKER CURRENT ADJUST”.
4. Illustration 3-1 shows the Input Voltage Selection terminal block. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



INPUT VOLTAGE SELECT TERMINAL BLOCK
Illustration 3-1

5. The overload and short circuit trip settings for the Main Input Circuit Breaker must be set to the correct values corresponding to the input voltage.

    5a. Remove the small cover plate under the Main Input Circuit Breaker which is held in place by two screws and lift open the transparent plastic cover on the lower portion of the circuit breaker to give access to the dip switches controlling the circuit breaker operation.

    5b. Illustration 3-2 shows the position of the dip switches for each input voltage selection. The switches immediately under the letter “L” are the only ones that should be changed. All others must be left in their “as shipped” up position.

3-1.2 Connection

Note the Cable Access Panel shown in Illustration 3-6 is located at the lower portion of the rear of the PDU. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

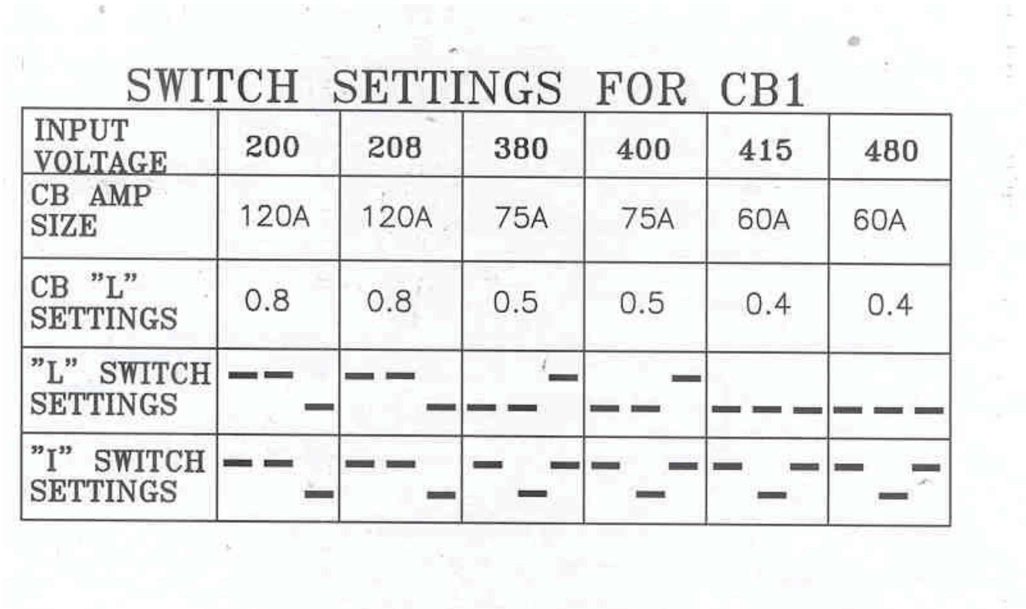
1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable Access Panel and tighten the cable clamps.

⚠ IMPORTANT

For input cables, use wire sizes in accordance with National Electric Code Table 310-16. The following table gives maximum full-load input currents in amperes (which would occur at 10% undervoltage) for each of the input voltages, as a guide to selecting proper wire sizes.

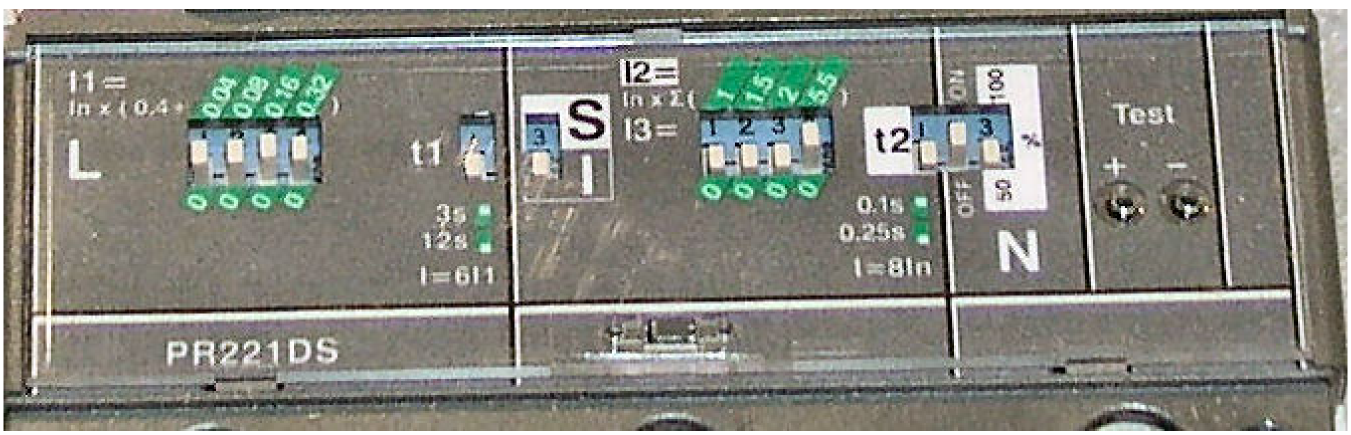
Maximum Input Current vs. Input Voltage

Table 3-1



SWITCH SETTINGS FOR CB1

INPUT VOLTAGE	200	208	380	400	415	480
CB AMP SIZE	120A	120A	75A	75A	60A	60A
CB "L" SETTINGS	0.8	0.8	0.5	0.5	0.4	0.4
"L" SWITCH SETTINGS						
"I" SWITCH SETTINGS						



CIRCUIT BREAKER DIP SWITCH SETTINGS
Illustration 3-2

3-2 Operation

3-2.1 Main Input Circuit Breaker

The Main Input Circuit Breaker operating handle has three positions:

- OFF (as shown in Illustration 3-4),
- ON (vertical), and
- TRIPPED (slightly to the right of vertical). If the Main Input Circuit Breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.

The Main Input Circuit Breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3-2.2 Power Off

The Power OFF button at the top of the front panel, when pressed, will immediately trip the Main Input Circuit Breaker, interrupting all power to the load.

3-2.3 EMO Reset

The EMO Reset button immediately to the right of the Power Off pushbutton will enable the Emergency Stop circuit.

3-2.4 TVSS Indicators

To the right of the Main Input Circuit Breaker are the TVSS indicators. A cutout makes visible the indicator on each of the three TVSS modules. Should a TVSS module fail because of a particularly large voltage surge, a red flag on the module indicator will appear and the Main Input Circuit Breaker will trip. Therefore, should the Main Circuit Breaker trip unexpectedly, check the TVSS Indicators for one or more failed modules showing the red flag. Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See Section 4-1.1, Replacement of a TVSS Module, for instructions for replacing a TVSS Module.

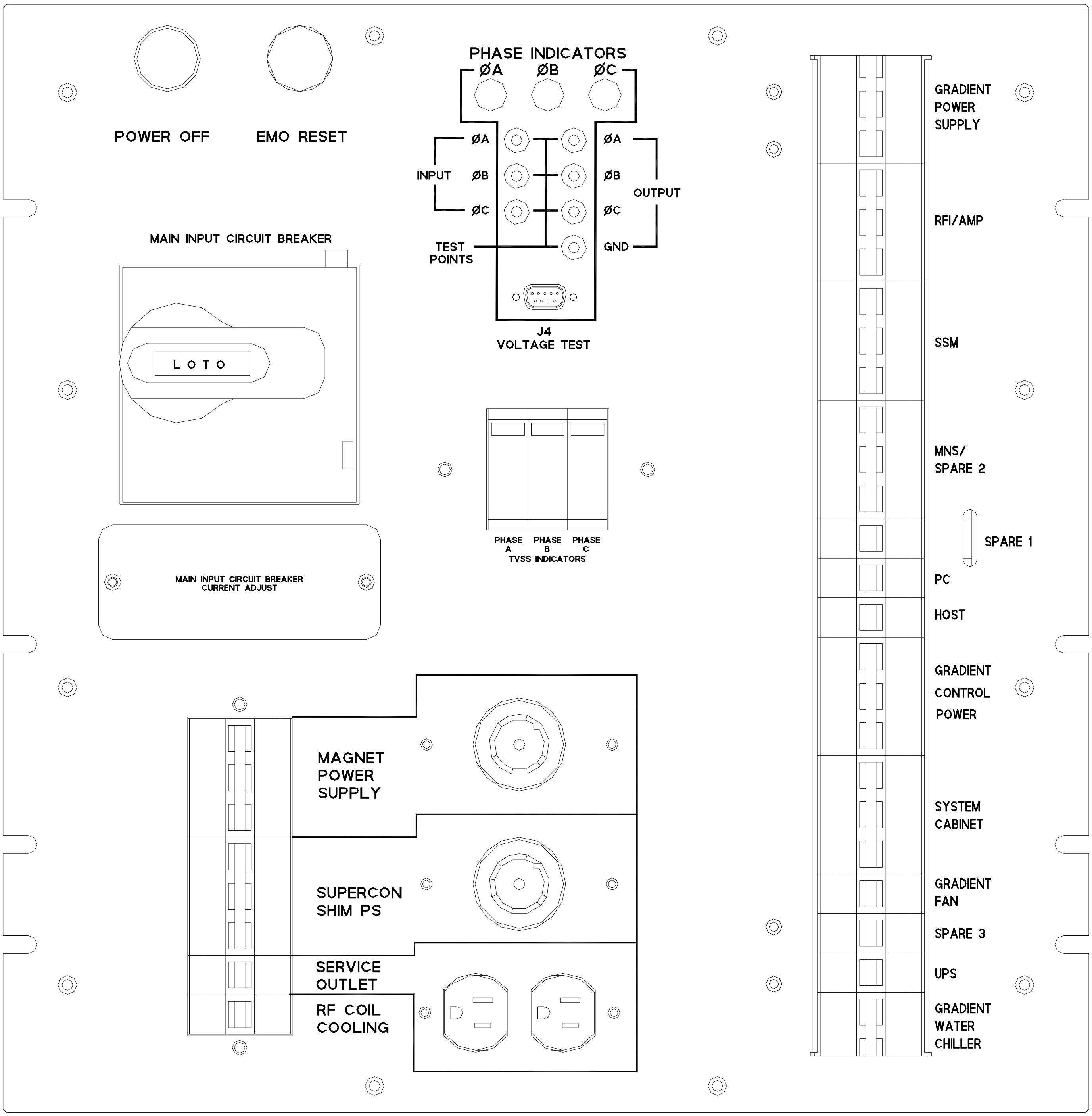
3-2.5 Phase Indicators

Just above the TVSS Indicators are three Phase Indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

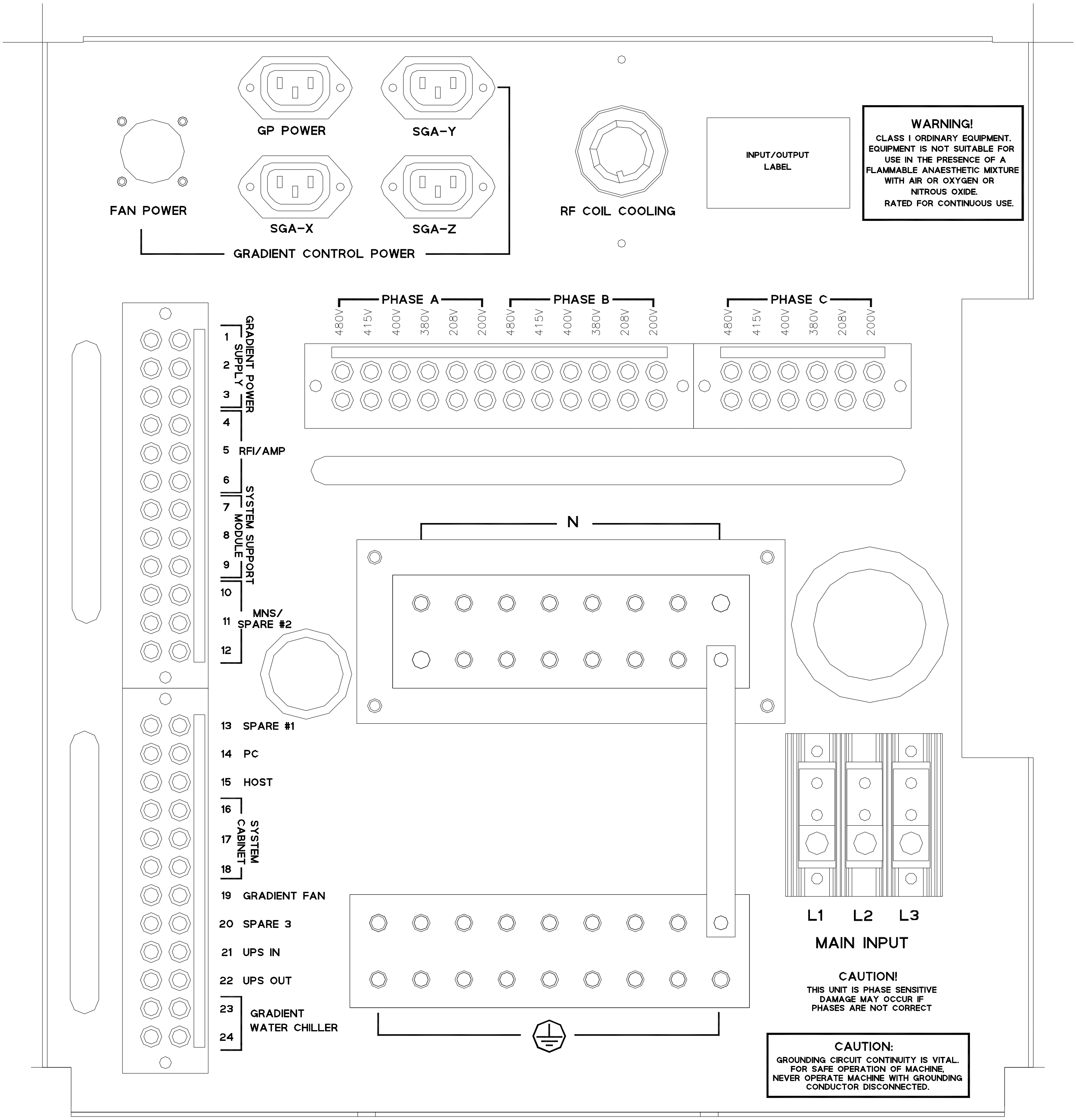
3-2.6 Input / Output Voltage Test Points

Immediately below the Phase Indicators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See Illustration 3-4. The voltages at these points are reduced by a factor of 100:1 from the actual voltage at the terminal blocks.

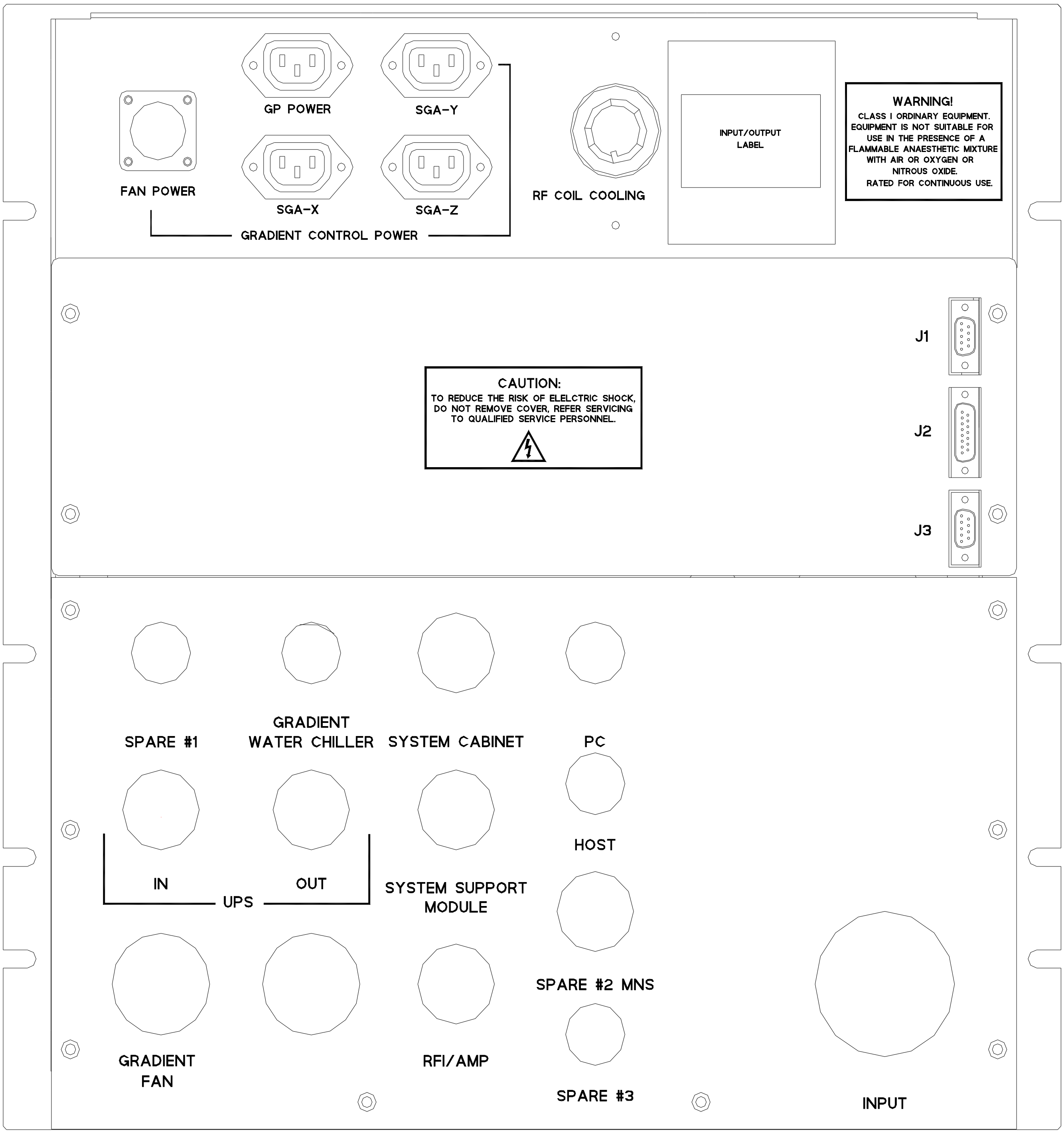
FRONT PANEL COMPONENT LOCATIONS
Illustration 3-4



REAR PANEL COMPONENT LOCATIONS
Illustration 3-5



REAR PANEL COMPONENT LOCATIONS
Illustration 3-6



3- FIELD REPLACEABLE UNITS (FRU'S)

LISTING OF FRU's
Table 4-1

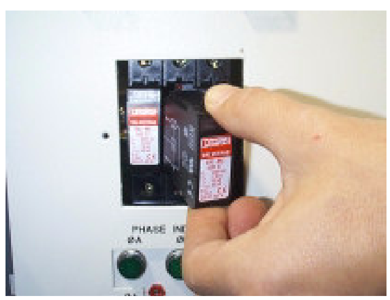
Item	Description	GEMS No.	Teal Part No.	FRU Type
1	TVSS Module	2301200-3	4990387	1
2	Control PCB	2321162-2	4990507	1

4-1 FRU Installation

4-1.1 Replacement of the TVSS Module

The operation of the TVSS Indicators is explained in Section 3-2.4, TVSS Indicators. If one (or more) of the indicators shows that a TVSS Module has failed, follow the steps below to replace the failed module and restore TVSS protection:

1. On the MDP, turn off the PDU circuit breaker, lock and tag.
2. Using a voltmeter or other voltage indicating device, check the voltage at the Main Input Terminal Board to be sure that no voltage is being applied to the input of the PDU. Main Input Terminal Board is located on rear panel (refer to illustration 3-5).
3. Pull out the failed TVSS Module(s) and plug in the replacement(s) as shown in Illustration 4-2.



TVSS MODULE REPLACEMENT
Illustration 4-2

Verify that all TVSS Modules are good (no red flags displayed).

4-1.2 Replacement of the Control PC Board

If the control board is deemed defective as per the Troubleshooting Guide in 4-1.3 follow the steps below to replace the control board. The control board is located at the back left side panel.

1. Turn off Main Input Circuit Breaker.
2. Lockout/Tagout the Main Input Circuit Breaker.
3. Remove the rear connector panel.
4. Remove control board from it's four (4) nylon standoff mounting locations.
5. Remove the connectors to the control board.
6. Connect the connectors removed in step 4. Back onto the new control board.
7. Install new control board at the four (4) nylon standoff mounting locations.
8. Install rear connector panel.
9. Remove Lockout/Tagout from Main Input Circuit Breaker.
10. Turn on main input circuit breaker.

Control PC Board, Internal View
From Rear of PDU
Illustration 4-3

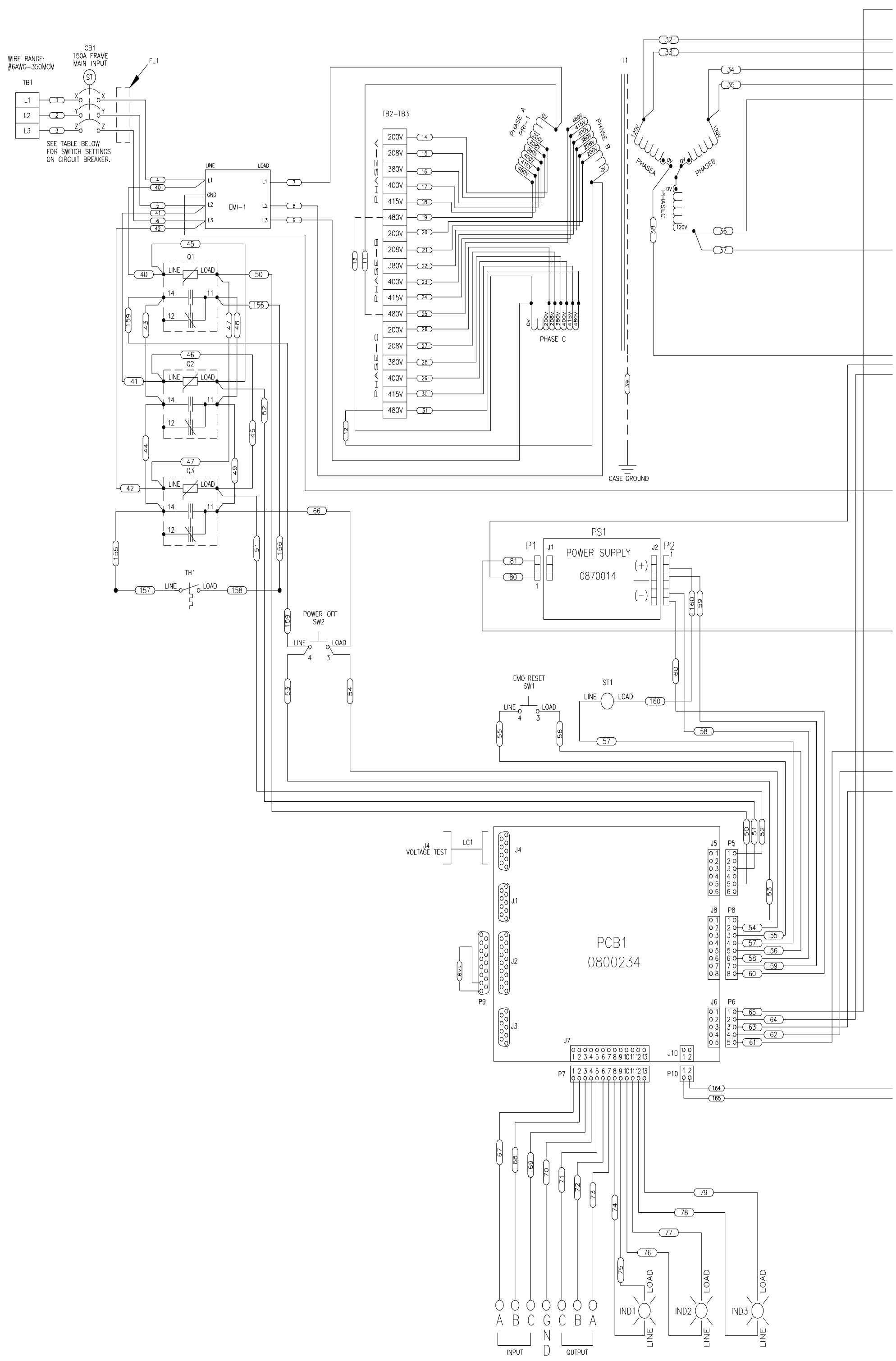
4-1.3 Troubleshooting Guide

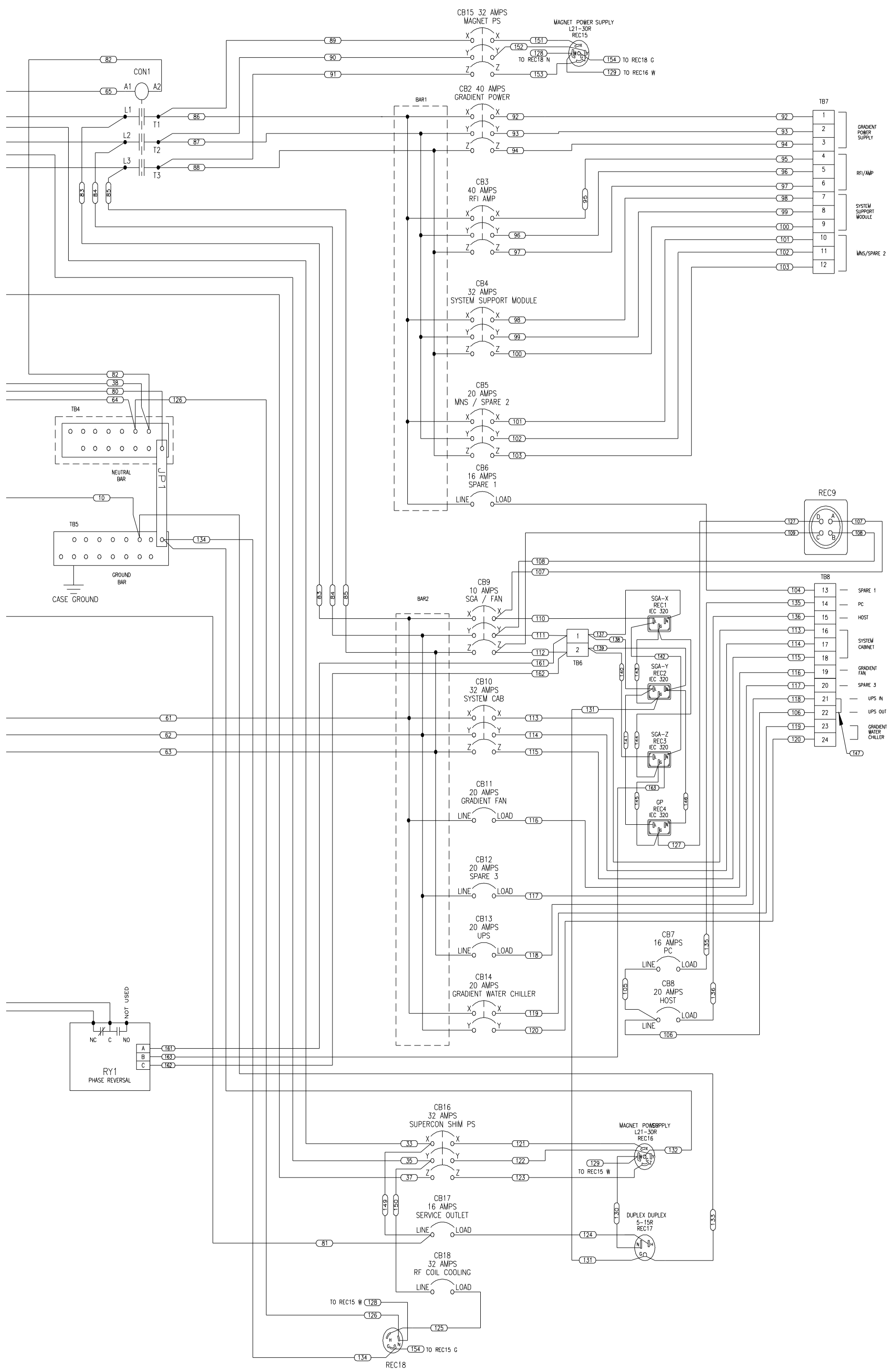
The following table lists fault conditions that may be encountered along with possible causes and solutions.

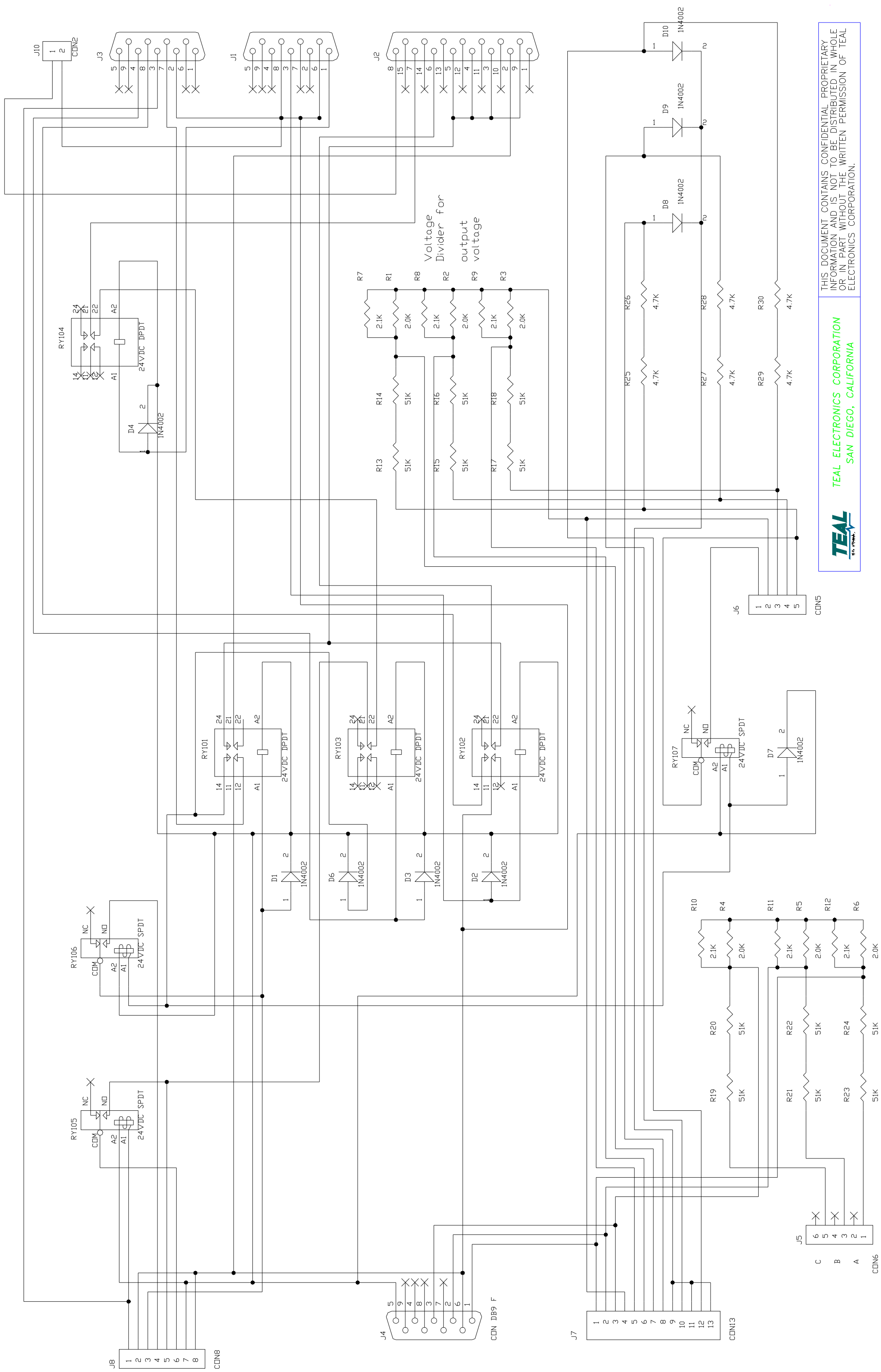
Troubleshoot Guide
Table 4-2

MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Main Input CB Trips	TVSS Module Failure	1. Check TVSS Modules for failure. A red "flag" will appear in TVSS module window when failure has occurred. 2. Replace defective TVSS module.
		Overtemperature	1. Investigate cause of overtemperature 2. Allow unit to cool down. 3. Reset the circuit breaker
		Shorted Power Off Switch	1. Turn power off 2. Reset Main Circuit Breaker 3. Turn power on 4. If CB trips, replace PDU
		Open from J3-2 to J3-3	1. Disconnect cable at J3 2. With DVM check for open at cable connector P3-2 to P3-3.
	Distribution Breakers Tripping	Defective Circuit Breaker	1. Turn power off 2. Check if circuit breaker still trips (won't latch up)
		Overload	1. Check load by disconnecting cabinet from PDU whose circuit breaker is tripping, or use an Amp Clamp / DVM to check current draw. 2. Correct load problem.
	CON1 Contactor not Operating	Defective Contactor	1. Ensure that the contactor CON1 is receiving 120V to the coils. 2. If 120V is present, replace PDU
		Defective Control Board PCB2	1. Disconnect cable at J2. 2. Check at cable connector if P2-7 to P2-8 is shorted. 3. If shorted, replace Control Board first. If trouble still remains, replace PDU 4. If not shorted, connect test plug and verify PDU operation. 5. If PDU operates with Test Plug, check the Emergency Loop outside of PDU. 6. If PDU does not operate with Test Plug, the trouble lies in both PDU and Emergency Loop outside of PDU (Possibly test plug too). Check and fix one by one.
		Incorrect Phasing	1. Turn power off to PDU 2. Re-connect input wires with proper phasing
	Phase Indicators not lighting when power is applied	Input Circuit Breaker off	1. Turn Input Circuit Breaker, CB1, on.
		Input Phase Missing	1. Check that other phase LED's are on. 2. Replace PDU
		Light Burned Out	1. Check that other phase LED's are on. 2. Replace PDU
		Not Getting Proper Output	1. Measure input and output, at front panel test points, of each phase, from phase to phase (208 VAC: 208 VAC) and phase to ground (120 VAC: 120 VAC). 2. If difference is not a 100:1 ratio, replace PDU

SCHEMATICS




CB15 32 AMPS MAGNET PS
MAGNET POWER SUPPLY L21-30R REC15
CON1
CB2 40 AMPS GRADIENT POWER
BAR1
TB7
GRADIENT POWER SUPPLY
RFI/AMP
SYSTEM SUPPORT MODULE
MNS/SPARE 2
CB3 40 AMPS RFI AMP
CB4 32 AMPS SYSTEM SUPPORT MODULE
CB5 20 AMPS MNS / SPARE 2
CB6 16 AMPS SPARE 1
TB4
NEUTRAL BAR
TB5
GROUND BAR
CASE GROUND
REC9
TB8
SPARE 1
PC
HOST
SYSTEM CABINET
GRADIENT FAN
SPARE 3
UPS IN
UPS OUT
GRADIENT WATER CHILLER
BAR2
CB9 10 AMPS SGA / FAN
TB6
SGA-X REC1 IEC 320
SGA-Y REC2 IEC 320
SGA-Z REC3 IEC 320
GP REC4 IEC 320
CB10 32 AMPS SYSTEM CAB
CB11 20 AMPS GRADIENT FAN
CB12 20 AMPS SPARE 3
CB13 20 AMPS UPS
CB14 20 AMPS GRADIENT WATER CHILLER
CB7 16 AMPS PC
CB8 20 AMPS HOST
NOT USED
RY1 PHASE REVERSAL
CB16 32 AMPS SUPERCON SHIM PS
MAGNET POWER SUPPLY L21-30R REC16
CB17 16 AMPS SERVICE OUTLET
DUPLEX DUPLEX 5-15R REC17
CB18 32 AMPS RF COIL COOLING
REC18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SAN DIEGO, CALIFORNIA
Voltage Divider for output voltage

Control PC Board, Internal View

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	1-13-03	TJD	New Release
0	7-29-03	TC	Production Release
1	2-2-07	RR	Revised 4-1.3 Troubleshooting Cause Column (Defective Control Board PCB2) items 3, 5 and 6.